SIGGRAPH 2001

Course 8

An Introduction to the Kalman Filter

Greg Welch
Gary Bishop

University of North Carolina at Chapel Hill
Department of Computer Science
Chapel Hill, NC 27599-3175

http://www.cs.unc.edu/~{welch, gb}
{welch, gb}@cs.unc.edu

©Copyright 2001 by ACM, Inc.
http://info.acm.org/pubs/toc/CRnotice.html

TABLE OF CONTENTS

TABLE OF CONTENTS 1
Preface 3
Course Syllabus 4

1. Introduction 5
    1.1 Course Description 5
    1.2 Speaker/Author Biographies 6

2. Probability and Random Variables 7
    2.1 Probability 7
    2.2 Random Variables 7
    2.3 Mean and Variance 9
    2.4 Normal or Gaussian Distribution 10
    2.5 Continuous Independence and Cond. Probability 11
    2.6 Spatial vs. Spectral Signal Characteristics 12

3. Stochastic Estimation 15
    3.1 State-Space Models 15
    3.2 The Observer Design Problem 16

4. The Kalman Filter 19
    4.1 The Discrete Kalman Filter 20
    4.2 The Extended Kalman Filter (EKF) 24
    4.3 An Example: Estimating a Random Constant 29

5. Other Topics 35
    5.1 Parameter Estimation or Tuning 35
    5.2 Multi-Modal (Multiple Model) Filters 36
    5.3 Hybrid or Multi-Sensor Fusion 40
    5.4 Single-Constraint-at-a-Time (SCAAT) 41

A. Bibliography 43

B. Related Papers 47

3. Stochastic Estimation

While there are many application-specific approaches to "computing" (estimating) an unknown state from a set of process measurements, many of these methods do not inherently take into consideration the typically *noisy* nature of the measurements. For example, consider our work in tracking for interactive computer graphics. While the requirements for the tracking information varies with application, the fundamental source of information is the same: pose estimates are derived from noisy electrical measurements of mechanical, inertial, optical, acoustic, or magnetic sensors. This noise is typically statistical in nature (or can be effectively modeled as such), which leads us to *stochastic* methods for addressing the problems. Here we provide a very basic introduction to the subject, primarily aimed at preparing the reader for Chapter 4. For a more extensive discussion of stochastic estimation see for example (Lewis 1986; Kailath, Sayed et al. 2000).

3.1 State-Space Models

State-space models are essentially a notational convenience for estimation and control problems, developed to make tractable what would otherwise be a notationally-intractable analysis. Consider a dynamic process described by an n-th order difference equation (similarly a differential equation) of the form

$$y_{i+1} = a_{0,i} y_i + \ldots + a_{n-1,i} y_{i-n+1} + u_i \,, \; i \geq 0 \,,$$

where $\{u_i\}$ is a *zero-mean* (statistically) *white* (spectrally) random "noise" process with autocorrelation

$$E(u_i, u_j) = R_u = Q_i \delta_{ij} \,,$$

and initial values $\{y_0, y_{-1}, \ldots, y_{-n+1}\}$ are zero-mean random variables with a known $n \times n$ *covariance matrix*

$$P_0 = E(y_{-j}, y_{-k}) \,, \; j, k \in \{0, n-1\} \,.$$

Also assume that

$$E(u_i, y_i) = 0 \text{ for } -n+1 \leq j \leq 0 \text{ and } i \geq 0 \,,$$

which ensures (Kailath, Sayed et al. 2000) that

$$E(u_i, y_i) = 0\,,\; i \geq j \geq 0\,.$$

In other words, that the noise is statistically independent from the process to be estimated. Under some other basic conditions (Kailath, Sayed et al. 2000) this difference equation can be re-written as

$$\hat{x}_{i+1} \equiv \begin{bmatrix} y_{i+1} \\ y_i \\ y_{i-1} \\ \vdots \\ y_{i-n+2} \end{bmatrix} = \underbrace{\begin{bmatrix} a_0 & a_1 & \dots & a_{n-2} & a_{n-1} \\ 1 & 0 & \dots & 0 & 0 \\ 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \dots & \vdots & \vdots \\ 0 & 0 & \dots & 1 & 0 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} y_i \\ y_{i-1} \\ y_{i-2} \\ \vdots \\ y_{i-n+1} \end{bmatrix}}_{\hat{x}_i} + \underbrace{\begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}}_{G} u_i$$

which leads to the *state-space model*

$$\hat{x}_{i+1} = A\hat{x}_i + Gu_i$$

$$\hat{y}_i = \begin{bmatrix} 1 & 0 & \dots & 0 \end{bmatrix} \hat{x}_i$$

or the more general form

$$\hat{x}_{i+1} = A\hat{x}_i + Gu_i \tag{3.1}$$

$$\hat{y}_i = H_i\hat{x}_i\,. \tag{3.2}$$

Equation (3.1) represents the way a new state $\hat{x}_{i+1}$ is modeled as a linear combination of both the previous state $\hat{x}_i$ and some *process noise* u_i. Equation (3.2) describes the way the process measurements or *observations* $\hat{y}_i$ are derived from the internal state $\hat{x}_i$. These two equations are often referred to respectively as the *process model* and the *measurement model*, and they serve as the basis for virtually all linear estimation methods, such as the *Kalman filter* described below.

3.2 The Observer Design Problem

There is a related general problem in the area of linear systems theory generally called the *observer design problem*. The basic problem is to determine (estimate) the internal *states* of a linear system, given access only to the system's *outputs*. (Access to the system's control inputs is also presumed, but we omit that aspect here. See for example (Kailath, Sayed et al. 2000) for more information.) This is akin to what people often think of as the "black box" problem where you have access to some signals coming from the box (the outputs) but you cannot directly observe what's inside.

The many approaches to this basic problem are typically based on the state-space model presented in the previous section. There is typically a *process model* that models the transformation of the process state. This can usually be represented as a linear stochastic difference equation similar to equation (3.1):

$$x_k = Ax_{k-1} + Bu_k + w_{k-1}. \tag{3.3}$$

In addition there is some form of *measurement model* that describes the relationship between the process state and the measurements. This can usually be represented with a linear expression similar to equation (3.2):

$$z_k = Hx_k + v_k. \tag{3.4}$$

The terms w_k and v_k are random variables representing the process and measurement noise respectively. Note that in equation (3.4) we changed the dependent variable to z_k instead of y_k as in equation (3.2). The rationale is to reinforce the notion that the measurements to not have to be of elements of the state specifically, but can be any linear combination of the state elements.

Measurement and Process Noise

We consider here the common case of noisy sensor measurements. There are many sources of noise in such measurements. For example, each type of sensor has fundamental limitations related to the associated physical medium, and when pushing the envelope of these limitations the signals are typically degraded. In addition, some amount of random electrical noise is added to the signal via the sensor and the electrical circuits. The time-varying ratio of "pure" signal to the electrical noise continuously affects the *quantity* and *quality* of the information. The result is that information obtained from any one sensor must be qualified as it is interpreted as part of an overall sequence of estimates, and analytical measurement models typically incorporate some notion of random measurement noise or uncertainty as shown above.

There is the additional problem that the actual state transform model is completely unknown. While we can make predictions over relatively short intervals using models based on recent state transforms, such predictions assume that the transforms are predictable, which is not always the case. The result is that like sensor information, ongoing estimates of the state must be qualified as they are combined with measurements in an overall sequence of estimates. In addition, process models typically incorporate some notion of random motion or uncertainty as shown above.

4. The Kalman Filter

Within the significant toolbox of mathematical tools that can be used for stochastic estimation from noisy sensor measurements, one of the most well-known and often-used tools is what is known as the *Kalman filter*. The Kalman filter is named after Rudolph E. Kalman, who in 1960 published his famous paper describing a recursive solution to the discrete-data linear filtering problem (Kalman 1960). A very "friendly" introduction to the general idea of the Kalman filter is offered in Chapter 1 of (Maybeck 1979)—which is available from the above Kalman filter web site, *and* we have included it (with permission) in this course pack. A more complete introductory discussion can be found in (Sorenson 1970), which also contains some interesting historical narrative. More extensive references include (Gelb 1974; Maybeck 1979; Lewis 1986; Jacobs 1993; Brown and Hwang 1996; Grewal and Andrews 2001). In addition, for many years we have maintained a web site dedicated to the Kalman filter. This site contains links to related work, papers, books, and even some software including a new Java-based *Kalman Filter Learning Tool*.

`http://www.cs.unc.edu/~welch/kalman/`

The Kalman filter is essentially a set of mathematical equations that implement a predictor-corrector type estimator that is *optimal* in the sense that it minimizes the estimated *error* covariance—when some presumed conditions are met. Since the time of its introduction, the *Kalman filter* has been the subject of extensive research and application, particularly in the area of autonomous or assisted navigation. This is likely due in large part to advances in digital computing that made the use of the filter practical, but also to the relative simplicity and robust nature of the filter itself. Rarely do the conditions necessary for optimality actually exist, and yet the filter apparently works well for many applications in spite of this situation.

Of particular note here, the Kalman filter has been used extensively for tracking in interactive computer graphics. We use a *single-constraint-at-a-time* Kalman filter (see Section 5.4 on page 41) in our HiBall Tracking System (Welch, Bishop et al. 1999; Welch, Bishop et al. 2001) which is commercially available from 3rdTech (3rdTech 2000). It has also been used for motion prediction (Azuma and Bishop 1994; Azuma 1995), and it is used for multi-sensor (inertial-acoustic) fusion in the commercial Constellation™ wide-area tracking system by Intersense (Foxlin, Harrington et al. 1998; Intersense 2000). See also (Fuchs (Foxlin) 1993; Van Pabst and Krekel 1993; Azarbayejani and Pentland 1994; Emura and Tachi 1994; Emura and Tachi 1994; Mazuryk and Gervautz 1995).

4.1 The Discrete Kalman Filter

This section describes the filter in its original formulation (Kalman 1960) where the measurements occur and the state is estimated at discrete points in time.

4.1.1 The Process to be Estimated

The Kalman filter addresses the general problem of trying to estimate the state $x \in \Re^n$ of a discrete-time controlled process that is governed by the linear stochastic difference equation

$$x_k = Ax_{k-1} + Bu_k + w_{k-1}, \tag{4.1}$$

with a measurement $z \in \Re^m$ that is

$$z_k = Hx_k + v_k. \tag{4.2}$$

The random variables w_k and v_k represent the process and measurement noise (respectively). They are assumed to be independent (of each other), white, and with normal probability distributions

$$p(w) \sim N(0, Q), \tag{4.3}$$

$$p(v) \sim N(0, R). \tag{4.4}$$

In practice, the *process noise covariance* Q and *measurement noise covariance* R matrices might change with each time step or measurement, however here we assume they are constant.

The $n \times n$ matrix A in the difference equation equation (4.1) relates the state at the previous time step $k-1$ to the state at the current step k, in the absence of either a driving function or process noise. Note that in practice A might change with each time step, but here we assume it is constant. The $n \times l$ matrix B relates the optional control input $u \in \Re^l$ to the state x. The $m \times n$ matrix H in the measurement equation equation (4.2) relates the state to the measurement z_k. In practice H might change with each time step or measurement, but here we assume it is constant.

4.1.2 The Computational Origins of the Filter

We define $\hat{x}_k^- \in \Re^n$ (note the "super minus") to be our *a priori* state estimate at step k given knowledge of the process prior to step k, and $\hat{x}_k \in \Re^n$ to be our *a posteriori* state estimate at step k given measurement z_k. We can then define *a priori* and *a posteriori* estimate errors as

$$e_k^- \equiv x_k - \hat{x}_k^-, \text{ and}$$

$$e_k \equiv x_k - \hat{x}_k.$$

The *a priori* estimate error covariance is then

$$P_k^- = E[e_k^- e_k^{-T}], \tag{4.5}$$

and the *a posteriori* estimate error covariance is

$$P_k = E[e_k e_k^T]. \tag{4.6}$$

In deriving the equations for the Kalman filter, we begin with the goal of finding an equation that computes an *a posteriori* state estimate $\hat{x}_k$ as a linear combination of an *a priori* estimate $\hat{x}_k^-$ and a weighted difference between an actual measurement z_k and a measurement prediction $H\hat{x}_k^-$ as shown below in equation (4.7). Some justification for equation (4.7) is given in "The Probabilistic Origins of the Filter" found below.

$$\hat{x}_k = \hat{x}_k^- + K(z_k - H\hat{x}_k^-) \tag{4.7}$$

The difference $(z_k - H\hat{x}_k^-)$ in equation (4.7) is called the measurement *innovation*, or the *residual*. The residual reflects the discrepancy between the predicted measurement $H\hat{x}_k^-$ and the actual measurement z_k. A residual of zero means that the two are in complete agreement.

The $n \times m$ matrix K in equation (4.7) is chosen to be the *gain* or *blending factor* that minimizes the *a posteriori* error covariance equation (4.6). This minimization can be accomplished by first substituting equation (4.7) into the above definition for e_k, substituting that into equation (4.6), performing the indicated expectations, taking the derivative of the trace of the result with respect to K, setting that result equal to zero, and then solving for K. For more details see (Maybeck 1979; Jacobs 1993; Brown and Hwang 1996). One form of the resulting K that minimizes equation (4.6) is given by[1]

$$\begin{aligned} K_k &= P_k^- H^T (H P_k^- H^T + R)^{-1} \\ &= \frac{P_k^- H^T}{H P_k^- H^T + R} \end{aligned} . \tag{4.8}$$

Looking at equation (4.8) we see that as the measurement error covariance R approaches zero, the gain K weights the residual more heavily. Specifically,

$$\lim_{R_k \to 0} K_k = H^{-1}.$$

1. All of the Kalman filter equations can be algebraically manipulated into to several forms. Equation equation (4.8) represents the Kalman gain in one popular form.

On the other hand, as the *a priori* estimate error covariance P_k^- approaches zero, the gain K weights the residual less heavily. Specifically,

$$\lim_{P_k^- \to 0} K_k = 0 .$$

Another way of thinking about the weighting by K is that as the measurement error covariance R approaches zero, the actual measurement z_k is "trusted" more and more, while the predicted measurement $H\hat{x}_k^-$ is trusted less and less. On the other hand, as the *a priori* estimate error covariance P_k^- approaches zero the actual measurement z_k is trusted less and less, while the predicted measurement $H\hat{x}_k^-$ is trusted more and more.

4.1.3 The Probabilistic Origins of the Filter

The justification for equation (4.7) is rooted in the probability of the *a priori* estimate $\hat{x}_k^-$ conditioned on all prior measurements z_k (Bayes' rule). For now let it suffice to point out that the Kalman filter maintains the first two moments of the state distribution,

$$E[x_k] = \hat{x}_k$$

$$E[(x_k - \hat{x}_k)(x_k - \hat{x}_k)^T] = P_k .$$

The *a posteriori* state estimate equation (4.7) reflects the mean (the first moment) of the state distribution— it is normally distributed if the conditions of equation (4.3) and equation (4.4) are met. The *a posteriori* estimate error covariance equation (4.6) reflects the variance of the state distribution (the second non-central moment). In other words,

$$\begin{aligned} p(x_k | z_k) &\sim N(E[x_k], E[(x_k - \hat{x}_k)(x_k - \hat{x}_k)^T]) \\ &= N(\hat{x}_k, P_k) . \end{aligned}$$

For more details on the probabilistic origins of the Kalman filter, see (Brown and Hwang 1996).

4.1.4 The Discrete Kalman Filter Algorithm

We will begin this section with a broad overview, covering the "high-level" operation of one form of the discrete Kalman filter (see the previous footnote). After presenting this high-level view, we will narrow the focus to the specific equations and their use in this version of the filter.

The Kalman filter estimates a process by using a form of feedback control: the filter estimates the process state at some time and then obtains feedback in the form of (noisy) measurements. As such, the equations for the Kalman filter fall into two groups: *time update* equations and *measurement update* equations. The time update equations are responsible for projecting forward (in time) the current state and error covariance

estimates to obtain the *a priori* estimates for the next time step. The measurement update equations are responsible for the feedback—i.e. for incorporating a new measurement into the *a priori* estimate to obtain an improved *a posteriori* estimate.

The time update equations can also be thought of as *predictor* equations, while the measurement update equations can be thought of as *corrector* equations. Indeed the final estimation algorithm resembles that of a *predictor-corrector* algorithm for solving numerical problems as shown below in Figure 4.1.

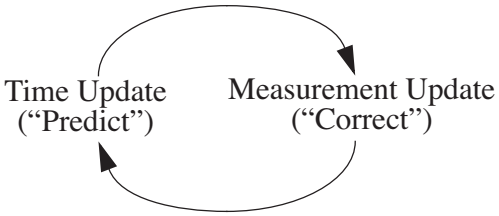


Figure 4.1: The ongoing discrete Kalman filter cycle. The *time update* projects the current state estimate ahead in time. The *measurement update* adjusts the projected estimate by an actual measurement at that time.

The specific equations for the time and measurement updates are presented below in table 4.1 and table 4.2.

Table 4.1: Discrete Kalman filter time update equations.

$$\hat{x}_k^- = A\hat{x}_{k-1} + Bu_k \tag{4.9}$$

$$P_k^- = AP_{k-1}A^T + Q \tag{4.10}$$

Again notice how the time update equations in table 4.1 project the state and covariance estimates forward from time step $k-1$ to step k. A and B are from equation (4.1), while Q is from equation (4.3). Initial conditions for the filter are discussed in the earlier references.

Table 4.2: Discrete Kalman filter measurement update equations.

$$K_k = P_k^- H^T (HP_k^- H^T + R)^{-1} \tag{4.11}$$

$$\hat{x}_k = \hat{x}_k^- + K_k(z_k - H\hat{x}_k^-) \tag{4.12}$$

$$P_k = (I - K_k H)P_k^- \tag{4.13}$$

The first task during the measurement update is to compute the Kalman gain, K_k. Notice that the equation given here as equation (4.11) is the same as equation (4.8). The next step is to actually measure the process to obtain z_k, and then to generate an *a posteriori* state estimate by incorporating the measurement as in equation (4.12). Again equation (4.12) is simply equation (4.7) repeated here for completeness. The final step is to obtain an *a posteriori* error covariance estimate via equation (4.13).

After each time and measurement update pair, the process is repeated with the previous *a posteriori* estimates used to project or predict the new *a priori* estimates. This recursive nature is one of the very appealing features of the Kalman filter—it makes practical implementations much more feasible than (for example) an implementation of a Wiener filter (Brown and Hwang 1996) which is designed to operate on *all* of the data *directly* for each estimate. The Kalman filter instead recursively conditions the current estimate on all of the past measurements. Figure 4.2 below offers a complete picture of the operation of the filter, combining the high-level diagram of Figure 4.1 with the equations from table 4.1 and table 4.2.

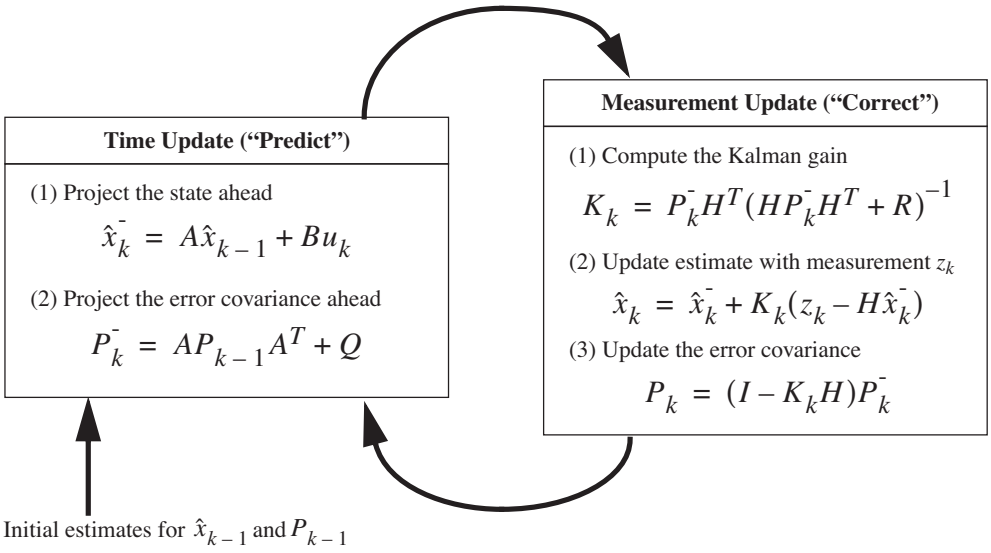


Figure 4.2: A complete picture of the operation of the Kalman filter, combining the high-level diagram of Figure 4.1 with the equations from table 4.1 and table 4.2.

4.2 The Extended Kalman Filter (EKF)

4.2.1 The Process to be Estimated

As described above in Section 4.1.1, the Kalman filter addresses the general problem of trying to estimate the state $x \in \Re^n$ of a discrete-time controlled process that is governed by a *linear* stochastic difference equation. But what happens if the process to be estimated and (or) the measurement relationship to the process is non-linear? Some of the most

interesting and successful applications of Kalman filtering have been such situations. A Kalman filter that linearizes about the current mean and covariance is referred to as an *extended Kalman filter* or EKF.

In something akin to a Taylor series, we can linearize the estimation around the current estimate using the partial derivatives of the process and measurement functions to compute estimates even in the face of non-linear relationships. To do so, we must begin by modifying some of the material presented in Section 4.1. Let us assume that our process again has a state vector $x \in \Re^n$, but that the process is now governed by the *non-linear* stochastic difference equation

$$x_k = f(x_{k-1}, u_k, w_{k-1}), \tag{4.14}$$

with a measurement $z \in \Re^m$ that is

$$z_k = h(x_k, v_k), \tag{4.15}$$

where the random variables w_k and v_k again represent the process and measurement noise as in equation (4.3) and equation (4.4). In this case the *non-linear* function f in the difference equation equation (4.14) relates the state at the previous time step $k-1$ to the state at the current time step k. It includes as parameters any driving function u_k and the zero-mean process noise w_k. The *non-linear* function h in the measurement equation equation (4.15) relates the state x_k to the measurement z_k.

In practice of course one does not know the individual values of the noise w_k and v_k at each time step. However, one can approximate the state and measurement vector without them as

$$\tilde{x}_k = f(\hat{x}_{k-1}, u_k, 0) \tag{4.16}$$

and

$$\tilde{z}_k = h(\tilde{x}_k, 0), \tag{4.17}$$

where $\hat{x}_k$ is some *a posteriori* estimate of the state (from a previous time step k).

It is important to note that a fundamental flaw of the EKF is that the distributions (or densities in the continuous case) of the various random variables are no longer normal after undergoing their respective nonlinear transformations. The EKF is simply an *ad hoc* state estimator that only approximates the optimality of Bayes' rule by linearization. Some interesting work has been done by Julier et al. in developing a variation to the EKF, using methods that preserve the normal distributions throughout the non-linear transformations (Julier and Uhlmann 1996).

4.2.2 The Computational Origins of the Filter

To estimate a process with non-linear difference and measurement relationships, we begin by writing new governing equations that linearize an estimate about equation (4.16) and equation (4.17),

$$x_k \approx \tilde{x}_k + A(x_{k-1} - \hat{x}_{k-1}) + W w_{k-1}\,, \tag{4.18}$$

$$z_k \approx \tilde{z}_k + H(x_k - \tilde{x}_k) + V v_k\,. \tag{4.19}$$

where

- x_k and z_k are the actual state and measurement vectors,
- $\tilde{x}_k$ and $\tilde{z}_k$ are the approximate state and measurement vectors from equation (4.16) and equation (4.17),
- $\hat{x}_k$ is an *a posteriori* estimate of the state at step k,
- the random variables w_k and v_k represent the process and measurement noise as in equation (4.3) and equation (4.4).
- A is the Jacobian matrix of partial derivatives of f with respect to x, that is

$$A_{[i,\,j]} = \frac{\partial f_{[i]}}{\partial x_{[j]}}(\hat{x}_{k-1}, u_k, 0)\,,$$

- W is the Jacobian matrix of partial derivatives of f with respect to w,

$$W_{[i,\,j]} = \frac{\partial f_{[i]}}{\partial w_{[j]}}(\hat{x}_{k-1}, u_k, 0)\,,$$

- H is the Jacobian matrix of partial derivatives of h with respect to x,

$$H_{[i,\,j]} = \frac{\partial h_{[i]}}{\partial x_{[j]}}(\tilde{x}_k, 0)\,,$$

- V is the Jacobian matrix of partial derivatives of h with respect to v,

$$V_{[i,\,j]} = \frac{\partial h_{[i]}}{\partial v_{[j]}}(\tilde{x}_k, 0)\,.$$

Note that for simplicity in the notation we do not use the time step subscript k with the Jacobians A, W, H, and V, even though they are in fact different at each time step.

Now we define a new notation for the prediction error,

$$\tilde{e}_{x_k} \equiv x_k - \tilde{x}_k \,, \tag{4.20}$$

and the measurement residual,

$$\tilde{e}_{z_k} \equiv z_k - \tilde{z}_k \,. \tag{4.21}$$

Remember that in practice one does not have access to x_k in equation (4.20), it is the *actual* state vector, i.e. the quantity one is trying to estimate. On the other hand, one *does* have access to z_k in equation (4.21), it is the actual measurement that one is using to estimate x_k. Using equation (4.20) and equation (4.21) we can write governing equations for an *error process* as

$$\tilde{e}_{x_k} \approx A(x_{k-1} - \hat{x}_{k-1}) + \varepsilon_k \,, \tag{4.22}$$

$$\tilde{e}_{z_k} \approx H\tilde{e}_{x_k} + \eta_k \,, \tag{4.23}$$

where ε_k and η_k represent new independent random variables having zero mean and covariance matrices WQW^T and VRV^T, with Q and R as in (4.3) and (4.4) respectively.

Notice that the equations equation (4.22) and equation (4.23) are linear, and that they closely resemble the difference and measurement equations equation (4.1) and equation (4.2) from the discrete Kalman filter. This motivates us to use the actual measurement residual $\tilde{e}_{z_k}$ in equation (4.21) and a second (hypothetical) Kalman filter to estimate the prediction error $\tilde{e}_{x_k}$ given by equation (4.22). This estimate, call it $\hat{e}_k$, could then be used along with equation (4.20) to obtain the *a posteriori* state estimates for the original non-linear process as

$$\hat{x}_k = \tilde{x}_k + \hat{e}_k \,. \tag{4.24}$$

The random variables of equation (4.22) and equation (4.23) have approximately the following probability distributions (see the previous footnote):

$$p(\tilde{e}_{x_k}) \sim N(0, E[\tilde{e}_{x_k}\tilde{e}_{x_k}^T])$$

$$p(\varepsilon_k) \sim N(0, WQ_kW^T)$$

$$p(\eta_k) \sim N(0, VR_kV^T)$$

Given these approximations and letting the predicted value of $\hat{e}_k$ be zero, the Kalman filter equation used to estimate $\hat{e}_k$ is

$$\hat{e}_k = K_k\tilde{e}_{z_k} \,. \tag{4.25}$$

By substituting equation (4.25) back into equation (4.24) and making use of equation (4.21) we see that we do not actually need the second (hypothetical) Kalman filter:

$$\begin{aligned} \hat{x}_k &= \tilde{x}_k + K_k \tilde{e}_{z_k} \\ &= \tilde{x}_k + K_k (z_k - \tilde{z}_k) \end{aligned} \tag{4.26}$$

Equation equation (4.26) can now be used for the measurement update in the extended Kalman filter, with $\tilde{x}_k$ and $\tilde{z}_k$ coming from equation (4.16) and equation (4.17), and the Kalman gain K_k coming from equation (4.11) with the appropriate substitution for the measurement error covariance.

The complete set of EKF equations is shown below in table 4.3 and table 4.4. Note that we have substituted $\hat{x}_k^-$ for $\tilde{x}_k$ to remain consistent with the earlier "super minus" a priori notation, and that we now attach the subscript k to the Jacobians A, W, H, and V, to reinforce the notion that they are different at (and therefore must be recomputed at) each time step.

Table 4.3: EKF time update equations.

$$\hat{x}_k^- = f(\hat{x}_{k-1}, u_k, 0) \tag{4.27}$$

$$P_k^- = A_k P_{k-1} A_k^T + W_k Q_{k-1} W_k^T \tag{4.28}$$

As with the basic discrete Kalman filter, the time update equations in table 4.3 project the state and covariance estimates from the previous time step $k-1$ to the current time step k. Again f in equation (4.27) comes from equation (4.16), A_k and W_k are the process Jacobians at step k, and Q_k is the process noise covariance equation (4.3) at step k.

Table 4.4: EKF measurement update equations.

$$K_k = P_k^- H_k^T (H_k P_k^- H_k^T + V_k R_k V_k^T)^{-1} \tag{4.29}$$

$$\hat{x}_k = \hat{x}_k^- + K_k (z_k - h(\hat{x}_k^-, 0)) \tag{4.30}$$

$$P_k = (I - K_k H_k) P_k^- \tag{4.31}$$

As with the basic discrete Kalman filter, the measurement update equations in table 4.4 correct the state and covariance estimates with the measurement z_k. Again h in equation (4.30) comes from equation (4.17), H_k and V are the measurement Jacobians at step k, and R_k is the measurement noise covariance equation (4.4) at step k. (Note we now subscript R allowing it to change with each measurement.)

The basic operation of the EKF is the same as the linear discrete Kalman filter as shown in Figure 4.1. Figure 4.3 below offers a complete picture of the operation of the EKF, combining the high-level diagram of Figure 4.1 with the equations from table 4.3 and table 4.4.

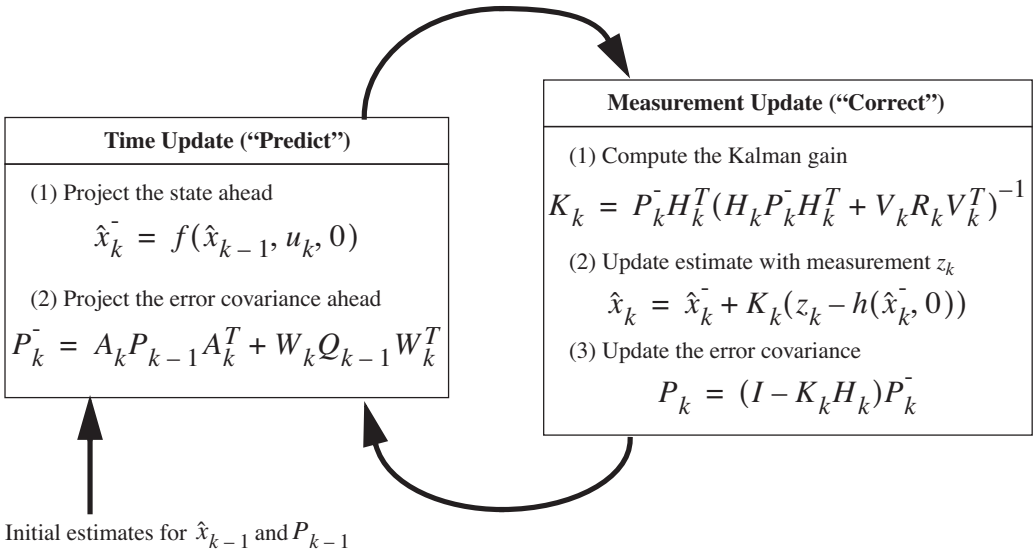


Figure 4.3: A complete picture of the operation of the *extended* Kalman filter, combining the high-level diagram of Figure 4.1 with the equations from table 4.3 and table 4.4.

An important feature of the EKF is that the Jacobian H_k in the equation for the Kalman gain K_k serves to correctly propagate or "magnify" only the relevant component of the measurement information. For example, if there is not a one-to-one mapping between the measurement z_k and the state via h, the Jacobian H_k affects the Kalman gain so that it only magnifies the portion of the residual $z_k - h(\hat{x}_k, 0)$ that does affect the state. Of course if over *all* measurements there is *not* a one-to-one mapping between the measurement z_k and the state via h, then as you might expect the filter will quickly diverge. In this case the process is *unobservable*.

4.3 An Example: Estimating a Random Constant

In the previous two sections we presented the basic form for the discrete Kalman filter, and the extended Kalman filter. To help in developing a better feel for the operation and capability of the filter, we present a very simple example here.

4.3.1 The Process Model

In this simple example let us attempt to estimate a scalar random constant, a voltage for example. Let's assume that we have the ability to take measurements of the constant, but that the measurements are corrupted by a 0.1 volt RMS *white* measurement noise (e.g. our analog to digital converter is not very accurate). In this example, our process is governed by the linear difference equation

$$\begin{aligned} x_k &= Ax_{k-1} + Bu_k + w_k \\ &= x_{k-1} + w_k \end{aligned},$$

with a measurement $z \in \Re^1$ that is

$$\begin{aligned} z_k &= Hx_k + v_k \\ &= x_k + v_k \end{aligned}.$$

The state does not change from step to step so $A = 1$. There is no control input so $u = 0$. Our noisy measurement is of the state directly so $H = 1$. (Notice that we dropped the subscript k in several places because the respective parameters remain constant in our simple model.)

4.3.2 The Filter Equations and Parameters

Our time update equations are

$$\hat{x}_k^- = \hat{x}_{k-1},$$

$$P_k^- = P_{k-1} + Q,$$

and our measurement update equations are

$$\begin{aligned} K_k &= P_k^-(P_k^- + R)^{-1} \\ &= \frac{P_k^-}{P_k^- + R} \end{aligned}, \tag{4.32}$$

$$\hat{x}_k = \hat{x}_k^- + K_k(z_k - \hat{x}_k^-),$$

$$P_k = (1 - K_k)P_k^-.$$

Presuming a very small process variance, we let $Q = 1e-5$. (We could certainly let $Q = 0$ but assuming a small but non-zero value gives us more flexibility in "tuning" the filter as we will demonstrate below.) Let's assume that from experience we know that the

true value of the random constant has a standard normal probability distribution, so we will "seed" our filter with the guess that the constant is 0. In other words, before starting we let $\hat{x}_{k-1} = 0$.

Similarly we need to choose an initial value for P_{k-1}, call it P_0. If we were absolutely certain that our initial state estimate $\hat{x}_0 = 0$ was correct, we would let $P_0 = 0$. However given the uncertainty in our initial estimate $\hat{x}_0$, choosing $P_0 = 0$ would cause the filter to initially and always believe $\hat{x}_k = 0$. As it turns out, the alternative choice is not critical. We could choose almost any $P_0 \neq 0$ and the filter would *eventually* converge. We'll start our filter with $P_0 = 1$.

4.3.3 The Simulations

To begin with, we randomly chose a scalar constant $z = -0.37727$ (there is no "hat" on the z because it represents the "truth"). We then simulated 50 distinct measurements z_k that had error normally distributed around zero with a standard deviation of 0.1 (remember we presumed that the measurements are corrupted by a 0.1 volt RMS *white* measurement noise). We could have generated the individual measurements within the filter loop, but pre-generating the set of 50 measurements allowed me to run several simulations with the same exact measurements (i.e. same measurement noise) so that comparisons between simulations with different parameters would be more meaningful.

In the first simulation we fixed the measurement variance at $R = (0.1)^2 = 0.01$. Because this is the "true" measurement error variance, we would expect the "best" performance in terms of balancing responsiveness and estimate variance. This will become more evident in the second and third simulation. Figure 4.4 depicts the results of this first simulation. The true value of the random constant $x = -0.37727$ is given by the solid line, the noisy measurements by the cross marks, and the filter estimate by the remaining curve.

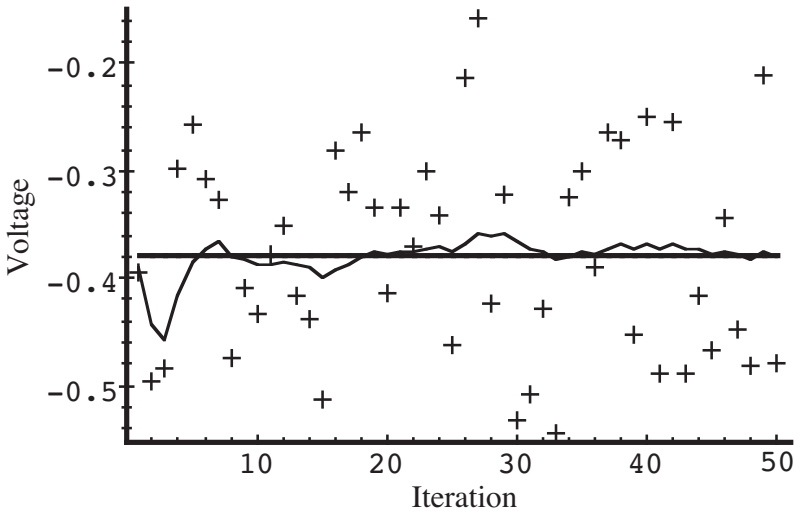


Figure 4.4: The first simulation: $R = (0.1)^2 = 0.01$. The true value of the random constant $x = -0.37727$ is given by the solid line, the noisy measurements by the cross marks, and the filter estimate by the remaining curve.

When considering the choice for P_0 above, we mentioned that the choice was not critical as long as $P_0 \neq 0$ because the filter would eventually converge. Below in Figure 4.5 we have plotted the value of P_k versus the iteration. By the 50th iteration, it has settled from the initial (rough) choice of 1 to approximately 0.0002 (Volts^2).

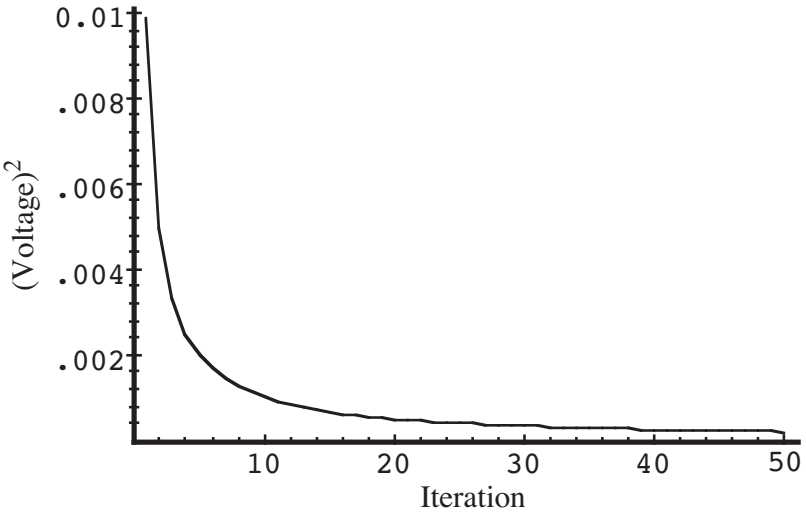


Figure 4.5: After 50 iterations, our initial (rough) error covariance P_k choice of 1 has settled to about 0.0002 (Volts^2).

In Section 5.1 under the topic "Parameter Estimation or Tuning" we briefly discussed changing or "tuning" the parameters Q and R to obtain different filter performance. In Figure 4.6 and Figure 4.7 below we can see what happens when R is increased or

decreased by a factor of 100 respectively. In Figure 4.6 the filter was told that the measurement variance was 100 times greater (i.e. $R = 1$) so it was "slower" to believe the measurements.

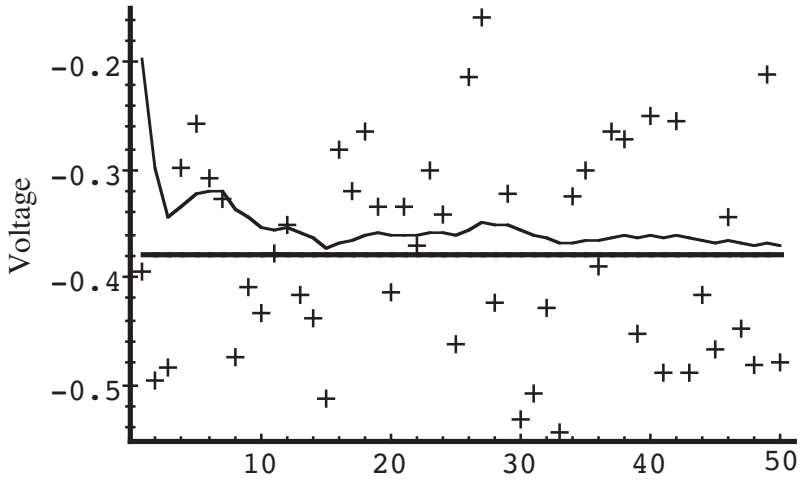


Figure 4.6: Second simulation: $R = 1$. The filter is slower to respond to the measurements, resulting in reduced estimate variance.

In Figure 4.7 the filter was told that the measurement variance was 100 times smaller (i.e. $R = 0.0001$) so it was very "quick" to believe the noisy measurements.

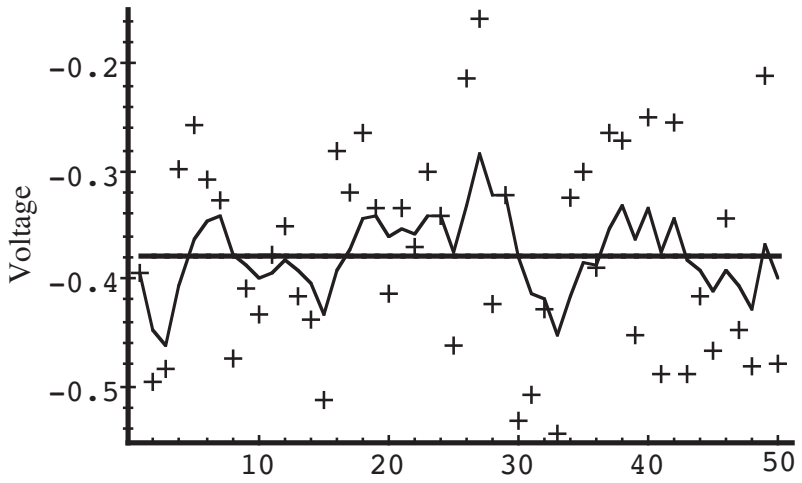


Figure 4.7: Third simulation: $R = 0.0001$. The filter responds to measurements quickly, increasing the estimate variance.

While the estimation of a constant is relatively straight-forward, it clearly demonstrates the workings of the Kalman filter. In Figure 4.6 in particular the Kalman "filtering" is evident as the estimate appears considerably smoother than the noisy measurements.